TRANSPORT AND TELECOMMUNICATION INSTITUTE

Investment group

MODERN DATABASE TECHNOLOGIES

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4403MDA

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Business case – Investment group

Introduction

The business case is about an investment group that owns several companies. The group's main business involves purchasing land in Latvia, primarily forestland, but the group also owns agricultural and other types of land. The group needs to build a database that will provide a list of all properties they own across the companies in the group, including assets held by each company. Additionally, the group aims to create queries for management to assist in decision-making and reporting.

1. Business Case Description

The investment group, which expands its activities in the territory of Latvia, deals with real estate, mostly buying up forest land.

The investment group includes several companies that belong exclusively to this group.

Each company has its own manager, but a manager can manage several enterprises of this group.

Enterprises does not have any limitation for how many real estate properties they have and will buy in future.

Each real estate property could be divided in to several land units and land unit could consist of 1- 3 types – forestland, agriculture land and other land.

1.1. Business objects:

- Company *(Comp_reg_no, Comp_name, Comp_Address, Comp_email)*
- Manager (*Mgr_pers_ID, Mgr_name , Mgr_surname, Mgr_email, Mgr_tel_No)*
- Real estate property *(Prop_ID, Prop_name,Prop_county, Prop_parish, Prop_purch_date, Prop_purch_value_EUR)*
- Land unit *(Unit_ID, Unit_cad_value_EUR)*
- Land *type (Type_ID, Type_area_ha)*

1.2. Business rules

One Group include many companies

One company belongs to one and only group

The company is managed by one manager

One manager could lead several companies

A company may own several real estate properties

One real estate property belongs to one company

One real estate property must include at least one land unit

One land unit should belong to one and only real estate property

One land unit consist of at least one, but up to three land types

One land type of unit belongs to one land unit

1.3. Possible requests that could be performed against the database:

a. Get a list off all properties?
b. How many hectares of land does the group own, divided by each land type?
c. How many real estate properties and hectares manages each manager?
d. Get a list of land units, where agricultural land is dominating?
e. Get a list of properties, where forestland is dominating?
f. How many EUR did each company spend on the purchase of real estate in the current and previous year?
g. Top 10 most expensive purchased properties EUR/ha in current year?

2. Database conceptual model

2.1. List of entities:

- Company *(Comp_reg_no, Comp_name, Comp_Address, Comp_email)*
- Manager (*Mgr_pers_ID, Mgr_name , Mgr_surname, Mgr_email, Mgr_tel_No)*
- Real estate property *(Prop_ID, Prop_name,Prop_county, Prop_parish, Prop_purch_date, Prop_purch_value_EUR)*
- Land unit *(Unit_ID, Unit_cad_value_EUR)*
- Land *type (Type_ID, Type_area_ha)*

2.2. Relationships:

The company is managed by one manager

One manager could lead several companies

- *Company : Manager -> M:1*

A company may own several real estate properties

One real estate property belongs to one company

- *Company : Property ->1:M*

One real estate property must include at least one land unit

One land unit should belong to one and only real estate property

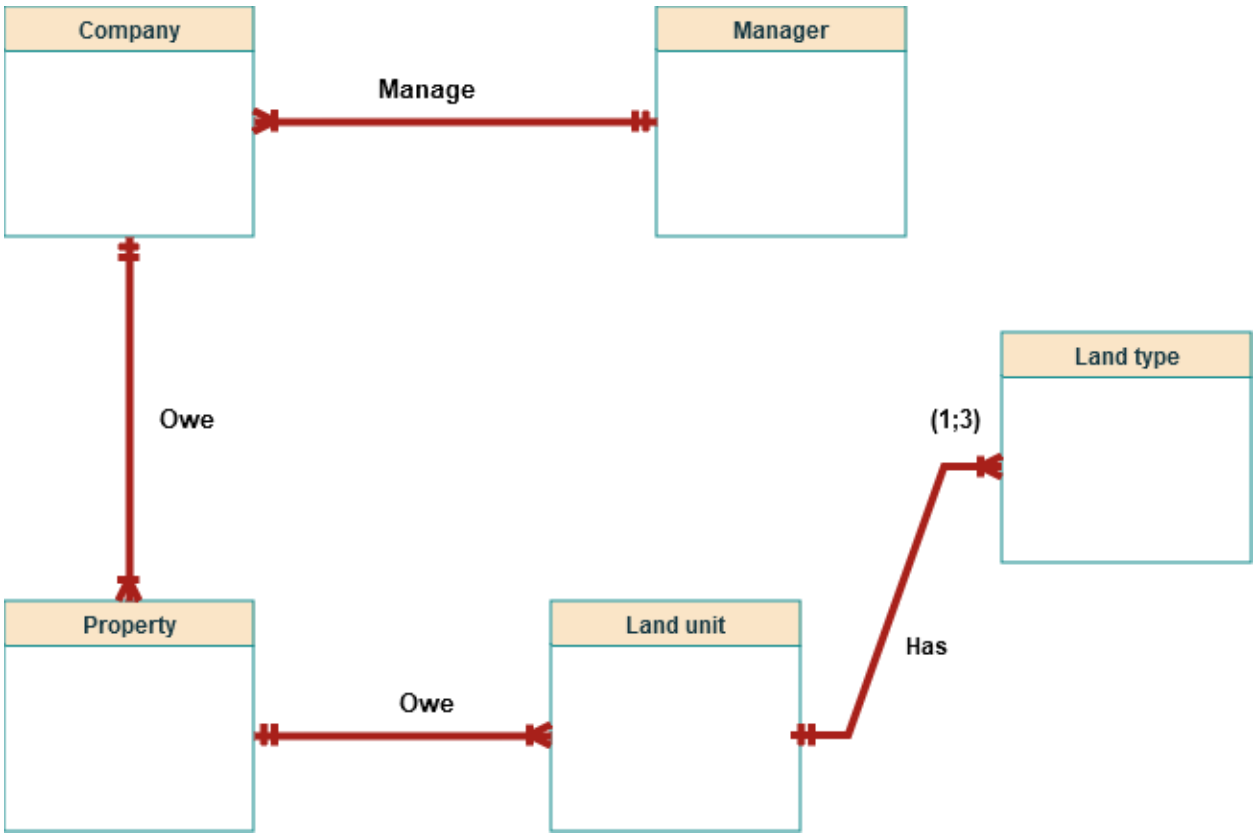
- *Property : Land unit ->1:M*

One land unit may consist of at least one land type

One land type of unit belongs to one land unit

- *Land unit : Land type -> 1:M*

2.3. ERD for the conceptual Model



2.4. Attributes and Primary Keys (Primary keys marked red)

- Company *(Comp_reg_no, Comp_name, Comp_Address, Comp_email)*
  PK
- Manager (*Mgr_pers_ID , Mgr _name, Mgr _surname, Mgr _email, Mgr _tel_No)*
  PK
- Poperty *(Prop_ID,Property_name, Prop_county, Prop_parish, Prop_purch_date, Prop_purch_value_EUR)*
  PK
- Land unit *(Unit_ID, Unit_cad_value_EUR)*
  PK
- Land type *(Type_ID, Type_area_ha)*

  PK

2.5. Attribute Value Domains (examples)

Entity	Attributes	Description	Data type, length	Null	Possible values, constraints, formats
Company	Comp_reg_no	Company registration number	Char(11)	No	XXXXXXXXXXX
	Comp_name	Company name	Varchar(100)	No	
	Comp_Address	Company address	Varchar(100)	Yes	
	Comp_email	Company email address	Varchar(30)	No	X…X@X…X
Manager	Mgr_pers_ID	Manager Personal ID number	Char(12)	No	XXXXXX-XXXXX
	Mgr_name	Manager name	Varchar(20)	No	
	Mgr_surname	Manager surname	Varchar(20)	No	
	Mgr_email	Manager email	Varchar(30)	No	X…X@X…X
	Mgr_tel_No	Manager phone number	Varchar(15)	Yes	
Real estate property	Prop_ID	Property cadaster number	Char(11)	No	XXXXXXXXXXX
	Prop_name	Property name	Varchar(20)	No	
	Prop_county	County	Varchar(20)	No	
	Prop_parish	Parish	Varchar(20)	Yes	
	Prop_purch_date	Property purchase date	Date	No	dd/mm/yyyy
	Prop_purch_value_EUR	Property purchase value(EUR)	Money	No	
Land unit	Unit_ID	Land unit identifier	Char(11)	No	XXXXXXXXXXX
	Unit_cad_value_EUR	Land plot cadarstral value	Money	No	
Land type	Type_ID	Land type ID	Char(12)	No	FXXXXXXXXXXX AXXXXXXXXXXX OXXXXXXXXXXX
	Type_area_ha	Land type area in ha	Decimal(10,2)	No	

2.6. Checking against queries: Do we have sufficient attributes?

a) Get a list off all properties?
We have property entity which contains – property ID, county, parish, area etc.

b) How many hectares of land does the group own, divided by each land type?
We have entity – land type and it contains land type attribute and area.

c) How many real estate properties and hectares manages each manager?
We have manager entity which is connected to company entity which is connected to real estate property entity, it contains area data.

d) Get a list of land units, where agricultural land is dominating?
There is a land type entity, which include type ID and it area in hectares, this entity is connected with entity land unit.

e) Get a list of properties, where forestland is dominating?
There is a land type entity, which include type ID and it area in hectares, this entity is connected with entity land unit and land unit entity is connected with property entity.

f) How many EUR did each company spend on the purchase of real estate in the current and previous year?
We have company entity, which is connected to real estate property, where is all necessary data-purchase date and purchase price.

g) Top 10 most expensive purchased properties EUR/ha in current year?
We can get data from property entity by using attributes – property price, property purchase date and land type entity – type area.

3. Logical model

3.1. Resolving the relationships

- **Company : Manager -> M:1**

Company*(Comp_reg_no, Mgr_pers_ID, Comp_name, Comp_Address, Comp_email)*

PK FK

Manager*(Mgr_pers_ID, Mgr_name, Mgr_surname, Mgr_email, Mgr_tel_No)*
PK

- **Company : Property ->1:M**

Company*(Comp_reg_no, Mgr_pers_ID, Comp_name, Comp_Address, Comp_email)*

PK FK

Property*(Prop_ID, Comp_reg_no, Prop_name, Prop_county, Prop_parish, Prop_purch_date, Prop_purch_value_EUR)*
PK FK

- **Property : Land unit ->1:M**

Property*((Prop_ID, Comp_reg_no, Prop_ name, Prop_county, Prop_parish, Prop_purch_date, Prop_purch_value_EUR)*

PK *FK*

Land unit(*Unit_ID, Prop_ID, Unit_cad_value_EUR)*

PK *FK*

- **Land unit : Land type -> 1:M**

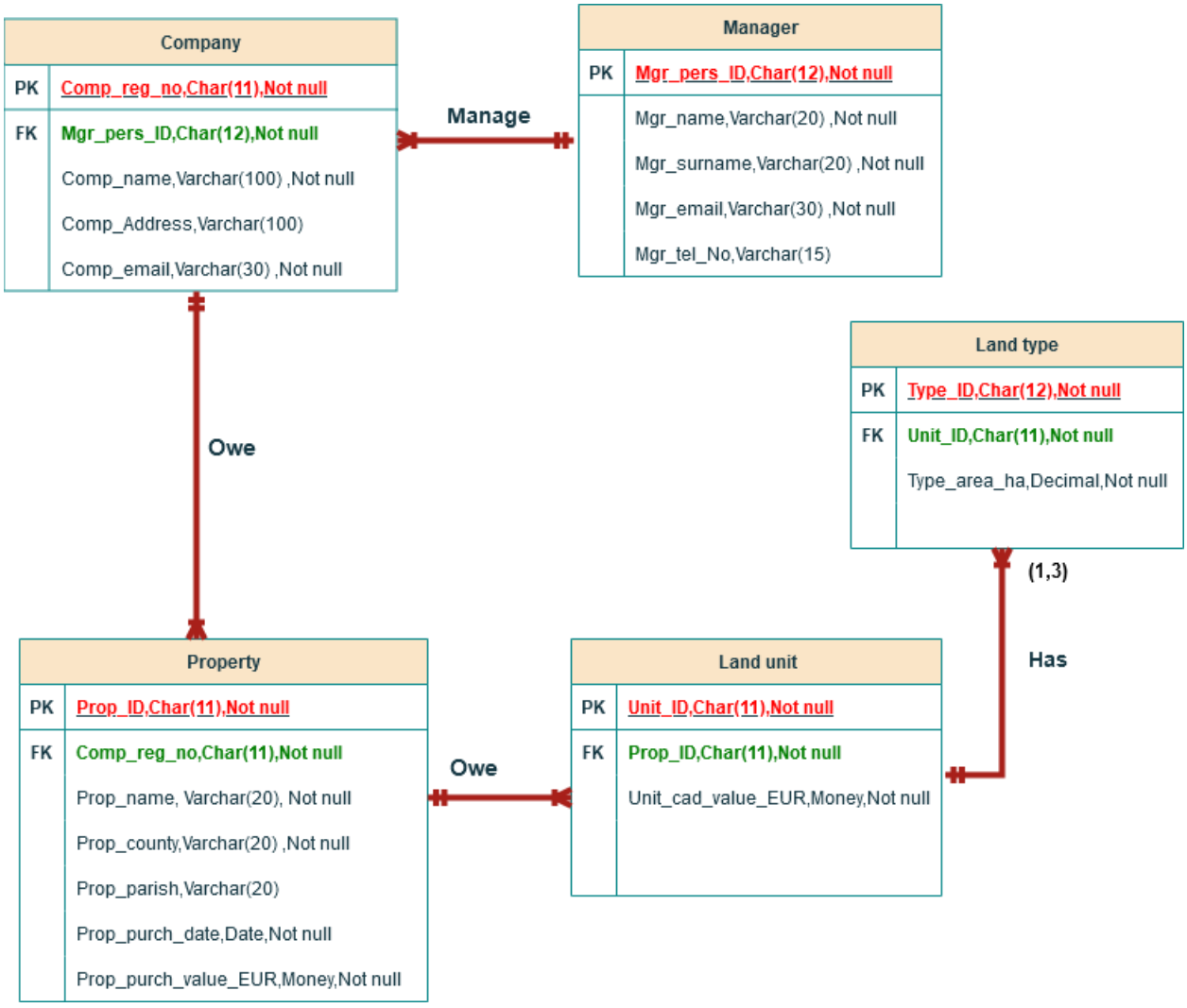
Land unit*(Unit_ID, Property_No, U_area_ha)*

PK *FK*

Land type*(Type_ID, Unit_ID ,Type_area_ha)*

PK *FK*

3.2. Logical model



3.3. Normalization justification

First Normal Form:

All primary keys are clearly defined;

All dependencies have been defined;

There are no repeating groups

Second Normal form:

All non-key attributes are fully functionally dependent on the primary key (no partial dependency).

Third Normal form:

There are no transitive dependencies (non-key attributes depending on other non-key attributes)

3.4. Brief description of the selected DBMS

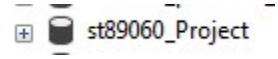
To effectively manage multiple records with their associated date ranges, a SQL relational database management system has been chosen as the solution for data storage and management.

Implementation

4. Creating database and tables

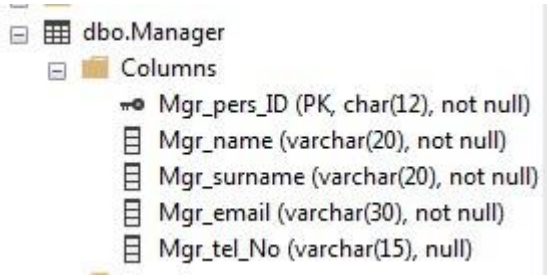
4.1. Creating database

```
create database st89060_Project
```



4.2. Creating table - Manager

```
create table Manager
(
Mgr_pers_ID Char(12) not null,
Mgr_name Varchar(20) not null,
Mgr_surname Varchar(20) not null,
Mgr_email Varchar(30) not null,
Mgr_tel_No Varchar(15),
Constraint C1 Primary key (Mgr_pers_ID)
)
```



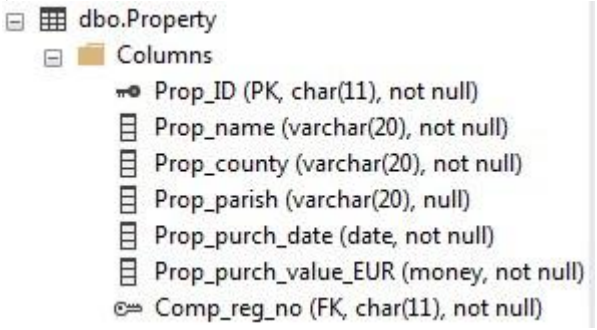
4.3. Creating table - Company

```
create table Company
(
Comp_reg_no char(11) not null,
Comp_name Varchar(100) not null,
Comp_Address Varchar(100),
Comp_email Varchar(30) not null,
Mgr_pers_ID Char(12) not null,
Constraint C2 Primary key (Comp_reg_no),
Constraint C3 Foreign key (Mgr_pers_ID) references Manager(Mgr_pers_ID)
)
```



4.4. Creating table - Property

```
create table Property
(
Prop_ID Char(11) not null,
Prop_name varchar(20) not null,
Prop_county Varchar(20) not null,
Prop_parish Varchar(20),
Prop_purch_date Date not null,
Prop_purch_value_EUR Money not null,
Comp_reg_no char(11) not null,
Constraint C4 Primary key (Prop_ID),
Constraint C5 Foreign key (Comp_reg_no) references Company(Comp_reg_no)
)
```



4.5. Creating table - Land unit

```
create table Land_unit
(
Unit_ID Char(11) not null,
Unit_cad_value_EUR Money not null,
Prop_ID Char(11) not null,
Constraint C6 Primary key (Unit_ID),
Constraint C7 Foreign key (Prop_ID) references Property(Prop_ID)
)
```



4.6. Creating table - Land Type

```
create table Land_type
(
Type_ID Char(12) not null,
Type_area_ha Decimal(10,2) not null,
Unit_ID Char(11) not null,
Constraint C8 Primary key (Type_ID),
Constraint C9 Foreign key (Unit_ID) references Land_unit(Unit_ID)
)
```



5. Inserting values

5.1. In to - Manager

```
insert into Manager
values
('291286-12315', 'Juris', 'Ezers', 'juris.ezers@gmail.com',null),
('150894-15675','Andris', 'Ozols', 'ozolos@inbox.lv','+37126712339'),
('240874-25684','Anna', 'Berga','berga@gmail.com',null)
```

	Mgr_pers_ID	Mgr_name	Mgr_surname	Mgr_email	Mgr_tel_No
1	150894-15675	Andris	Ozols	ozolos@inbox.lv	+37126712339
2	240874-25684	Anna	Berga	berga@gmail.com	NULL
3	291286-12315	Juris	Ezers	juris.ezers@gmail.com	NULL

5.2. In to – Company

```
insert into Company
values
('48502008094', 'Forest, I.K.', 'Lutriņi, Lutriņu pag., Saldus nov., LV-3861',
'forestry@gmail.com','291286-12315'),
('40103276250','SIA ECO FOREST',null, 'eco_forest@inbox.lv','150894-15675'),
('40103971103', 'SIA FORSH LAND', null, 'f_land123@inox.lv', '240874-25684'),
('40203471162', 'SIA Agro Meži', null, 'agromezi@gmail.com','150894-15675')
```

	Comp_reg_no	Comp_name	Comp_Address	Comp_email	Mgr_pers_ID
1	40103276250	SIA ECO FOREST	NULL	eco_forest@inbox.lv	150894-15675
2	40103971103	SIA FORSH LAND	NULL	f_land123@inox.lv	240874-25684
3	40203471162	SIA Agro Meži	NULL	agromezi@gmail.com	150894-15675
4	48502008094	Forest, I.K.	Lutrini, Lutrinu pag., Saldus nov., LV-3861	forestry@gmail.com	291286-12315

5.3. In to - Property

```
insert into Property
values
('32420090034','Austriņi','Aizkraukles','Aiviekstes','2024-08-18',88500,'48502008094'),
('32600020020','Bebrupīte','Aizkraukles','Kokneses','2024-03-15',38250,'48502008094'),
('36560060139','Jaunlīči','Siguldas','Mores','2022-01-22',16500,'40103276250'),
('36560060140','Lejas Līči','Siguldas','Mores','2023-05-23',63800,'40103276250'),
('40440080131','Jaungrantiņi','Gulbenes','Jaungulbenes','2023-07-
13',57585,'40103276250'),
('40640140210','Klīvi','Siguldas','Mores','2023-09-01',35200,'40103276250'),
('40920070168','Rūsas','Bauskas','Iecavas','2024-04-29',40700,'48502008094'),
('40940060025','Rūsa','Bauskas','Vecumnieku','2023-01-07',76230,'48502008094'),
('44500060057','Zemgales','Siguldas','Allažu','2024-01-22',21000,'40103971103'),
('44500060130','Bez nosaukuma','Augšdaugavas','Demenes','2023-05-
31',58000,'40103971103'),
('44500060197','Lakstīgalas 2','Siguldas','Allažu','2024-02-14',29040,'40103971103'),
('44720070076','Jāniši-3','Augšdaugavas','Medumu','2024-01-27',6520,'40103971103'),
('44940050403','Ganiņu mežs','Siguldas','Allažu','2024-11-23',11840,'40103971103'),
('50600010042','Jaunzemi','Gulbenes','Jaungulbenes','2022-12-12',69200,'40103971103'),
('50600020182','Jaunobravieši','Gulbenes','Jaungulbenes','2023-02-
16',332000,'40103971103'),
('54310030046','Mazpērlītes','Jelgavas','Kalnciema','2023-05-12',14440,'48502008094'),
('56350150027','Mežvidiņi','Krāslavas','Indras','2024-09-01',12980,'40103276250'),
('60580020248','Bebri','Madonas','Praulienas','2024-04-13',13360,'40103276250'),
('60620050056','Zaļmeži','Krāslavas','Indras','2022-03-15',12600,'40103276250'),
('60960020123','Lasenbergi','Krāslavas','Ūdrīšu','2024-09-15',23200,'48502008094'),
('70820070016','Lejas Boksti','Rēzeknes','Čornajas','2023-07-02',40400,'40203471162'),
('70860110069','Tursi','Madonas','Praulienas','2024-07-03',101150,'48502008094'),
('74920080027','Jauncelmiņi','Bauskas','Iecavas','2023-12-28',147050,'48502008094'),
```

```
('76860050185','Purvsala','Līvānu','Turku','2024-10-10',50150,'48502008094'),
('78460040396','Lazdiņi','Augšdaugavas','Demenes','2024-08-02',36125,'40203471162'),
('78780020272','Ežmalas','Gulbenes','Jaungulbenes','2024-08-07',585650,'40103276250'),
('80800190010','Vecrijnieki','Olaines','Olaines','2023-11-13',266050,'40103276250'),
('88420080043','Gundari','Siguldas','Allažu','2023-03-26',61200,'40203471162'),
('88420080082','Dāgas','Siguldas','Allažu','2023-12-21',79050,'40203471162'),
('88820030147','Māravas','Talsu','Rojas','2023-02-17',14450,'48502008094'),
('96700030090','Jaunmežiņi ','Rēzeknes','Čornajas','2023-01-19',198050,'40203471162');
```

	Prop_ID	Prop_name	Prop_county	Prop_parish	Prop_purch_date	Prop_purch_value_EUR	Comp_reg_no
1	32420090034	Austrini	Aizkraukles	Aiviekstes	2024-08-18	88500.00	48502008094
2	32600020020	Bebrupite	Aizkraukles	Kokneses	2024-03-15	38250.00	48502008094
3	36560060139	Jaunlici	Siguldas	Mores	2022-01-22	16500.00	40103276250
4	36560060140	Lejas Lici	Siguldas	Mores	2023-05-23	63800.00	40103276250
5	40440080131	Jaungrantini	Gulbenes	Jaungulbenes	2023-07-13	57585.00	40103276250
6	40640140210	Klivi	Siguldas	Mores	2023-09-01	35200.00	40103276250
7	40920070168	Rusas	Bauskas	Iecavas	2024-04-29	40700.00	48502008094
8	40940060025	Rusa	Bauskas	Vecumnieku	2023-01-07	76230.00	48502008094
9	44500060057	Zemgales	Siguldas	Allažu	2024-01-22	21000.00	40103971103
10	44500060130	Bez nosaukuma	Augšdaugavas	Demenes	2023-05-31	58000.00	40103971103
11	44500060197	Lakstigalas 2	Siguldas	Allažu	2024-02-14	29040.00	40103971103
12	44720070076	Janiši-3	Augšdaugavas	Medumu	2024-01-27	6520.00	40103971103
13	44940050403	Ganinu mežs	Siguldas	Allažu	2024-11-23	11840.00	40103971103
14	50600010042	Jaunzemi	Gulbenes	Jaungulbenes	2022-12-12	69200.00	40103971103
15	50600020182	Jaunobravieši	Gulbenes	Jaungulbenes	2023-02-16	332000.00	40103971103
16	54310030046	Mazperlites	Jelgavas	Kalnciema	2023-05-12	14440.00	48502008094
17	56350150027	Mežvidini	Kraslavas	Indras	2024-09-01	12980.00	40103276250
18	60580020248	Bebri	Madonas	Praulienas	2024-04-13	13360.00	40103276250
19	60620050056	Zalmeži	Kraslavas	Indras	2022-03-15	12600.00	40103276250
20	60960020123	Lasenbergi	Kraslavas	Udrišu	2024-09-15	23200.00	48502008094
21	70820070016	Lejas Boksti	Rezeknes	Comajas	2023-07-02	40400.00	40203471162
22	70860110069	Tursi	Madonas	Praulienas	2024-07-03	101150.00	48502008094
23	74920080027	Jauncelmini	Bauskas	Iecavas	2023-12-28	147050.00	48502008094
24	76860050185	Purvsala	Livanu	Turku	2024-10-10	50150.00	48502008094
25	78460040396	Lazdini	Augšdaugavas	Demenes	2024-08-02	36125.00	40203471162
26	78780020272	Ežmalas	Gulbenes	Jaungulbenes	2024-08-07	585650.00	40103276250
27	80800190010	Vecrijnieki	Olaines	Olaines	2023-11-13	266050.00	40103276250
28	88420080043	Gundari	Siguldas	Allažu	2023-03-26	61200.00	40203471162
29	88420080082	Dagas	Siguldas	Allažu	2023-12-21	79050.00	40203471162
30	88820030147	Maravas	Talsu	Rojas	2023-02-17	14450.00	48502008094
31	96700030090	Jaunmežini	Rezeknes	Comajas	2023-01-19	198050.00	40203471162

5.4. In to -Land unit

```
insert into Land_unit
values
('36560060055',394,'36560060139'),
('36560060104',5583,'36560060140'),
('40440080130',6395,'40440080131'),
('40640140211',255,'40640140210'),
('56350150027',1872,'56350150027'),
('60580020284',581,'60580020248'),
('60620050199',1609,'60620050056'),
('60620050056',125,'60620050056'),
('78780020272',6071,'78780020272'),
('80800190011',10438,'80800190010'),
('80800190010',9856,'80800190010'),
('44500060057',4168,'44500060057'),
('44500060022',3110,'44500060130'),
('44500060198',125,'44500060197'),
('44500060197',785,'44500060197'),
('44720070076',264,'44720070076'),
('44940050325',222,'44940050403'),
('50600010035',7430,'50600010042'),
('50600050089',1266,'50600020182'),
('50600010034',27686,'50600020182'),
('50600030153',12555,'50600020182'),
('70820080009',244,'70820070016'),
('70820080011',255,'70820070016'),
('70820070016',6394,'70820070016'),
('78460070360',525,'78460040396'),
('88420080043',3556,'88420080043'),
('88420080082',1679,'88420080082'),
('96700030090',9755,'96700030090'),
('32420090036',4665,'32420090034'),
('32420090035',9141,'32420090034'),
('32600020020',638,'32600020020'),
('40920070168',2123,'40920070168'),
('40940060025',4596,'40940060025'),
('40940070046',4566,'40940060025'),
('54310030114',1049,'54310030046'),
('60960030097',3306,'60960020123'),
('70860110069',4672,'70860110069'),
('74920080027',6541,'74920080027'),
('76860050185',3243,'76860050185'),
('88820030147',1133,'88820030147')
```

	Unit_ID	Unit_cad_value_EUR	Prop_ID
1	32420090035	9141.00	32420090034
2	32420090036	4665.00	32420090034
3	32600020020	638.00	32600020020
4	36560060055	394.00	36560060139
5	36560060104	5583.00	36560060140
6	40440080130	6395.00	40440080131
7	40640140211	255.00	40640140210
8	40920070168	2123.00	40920070168
9	40940060025	4596.00	40940060025
10	40940070046	4566.00	40940060025
11	44500060022	3110.00	44500060130
12	44500060057	4168.00	44500060057
13	44500060197	785.00	44500060197
14	44500060198	125.00	44500060197
15	44720070076	264.00	44720070076
16	44940050325	222.00	44940050403
17	50600010034	27686.00	50600020182
18	50600010035	7430.00	50600010042
19	50600030153	12555.00	50600020182
20	50600050089	1266.00	50600020182
21	54310030114	1049.00	54310030046
22	56350150027	1872.00	56350150027
23	60580020284	581.00	60580020248
24	60620050056	125.00	60620050056
25	60620050199	1609.00	60620050056
26	60960030097	3306.00	60960020123
27	70820070016	6394.00	70820070016
28	70820080009	244.00	70820070016
29	70820080011	255.00	70820070016
30	70860110069	4672.00	70860110069
31	74920080027	6541.00	74920080027
32	76860050185	3243.00	76860050185
33	78460070360	525.00	78460040396
34	78780020272	6071.00	78780020272
35	80800190010	9856.00	80800190010
36	80800190011	10438.00	80800190010
37	88420080043	3556.00	88420080043
38	88420080082	1679.00	88420080082
39	88820030147	1133.00	88820030147
40	96700030090	9755.00	96700030090

5.5. In to - Land type

```
insert into Land_type
values
('F36560060055',2.2,'36560060055'),
('F36560060104',11.6,'36560060104'),
('F40440080130',10.47,'40440080130'),
('A40640140211',6.4,'40640140211'),
('A56350150027',6.49,'56350150027'),
('A60580020284',6.68,'60580020284'),
('F60620050199',2.8,'60620050199'),
('F60620050056',3.5,'60620050056'),
('F78780020272',51.2,'78780020272'),
('A78780020272',16.5,'78780020272'),
('O78780020272',1.2,'78780020272'),
('F80800190011',5.8,'80800190011'),
('F80800190010',25.5,'80800190010'),
('F44500060057',4.96,'44500060057'),
('F44500060022',14.5,'44500060022'),
('F44500060198',2.29,'44500060198'),
('O44500060197',4.97,'44500060197'),
('F44720070076',1.63,'44720070076'),
('F44940050325',2.96,'44940050325'),
('F50600010035',17.3,'50600010035'),
('F50600050089',13.4,'50600050089'),
('F50600010034',14,'50600010034'),
('A50600010034',3,'50600010034'),
('F50600030153',47.2,'50600030153'),
('A50600030153',2.2,'50600030153'),
('O50600030153',3.2,'50600030153'),
('A70820080009',2.1,'70820080009'),
('A70820080011',5.6,'70820080011'),
('A70820070016',12.5,'70820070016'),
('F78460070360',4.25,'78460070360'),
('A88420080043',7.2,'88420080043'),
('A88420080082',9.3,'88420080082'),
('A96700030090',23.3,'96700030090'),
('F32420090036',1.3,'32420090036'),
('F32420090035',10.5,'32420090035'),
('F32600020020',5.1,'32600020020'),
('F40920070168',7.4,'40920070168'),
('F40940060025',6.8,'40940060025'),
('A40940070046',7.06,'40940070046'),
('A54310030114',7.22,'54310030114'),
('A60960030097',11.6,'60960030097'),
('F70860110069',11.9,'70860110069'),
('F74920080027',17.3,'74920080027'),
('F76860050185',5.9,'76860050185'),
('O88820030147',1.7,'88820030147')
```

--Totally affected 45 rows

	Type_ID	Type_area_ha	Unit_ID
1	A40640140211	6.40	40640140211
2	A40940070046	7.06	40940070046
3	A50600010034	3.00	50600010034
4	A50600030153	2.20	50600030153
5	A54310030114	7.22	54310030114
6	A56350150027	6.49	56350150027
7	A60580020284	6.68	60580020284
8	A60960030097	11.60	60960030097
9	A70820070016	12.50	70820070016
10	A70820080009	2.10	70820080009
11	A70820080011	5.60	70820080011
12	A78780020272	16.50	78780020272
13	A88420080043	7.20	88420080043
14	A88420080082	9.30	88420080082
15	A96700030090	23.30	96700030090
16	F32420090035	10.50	32420090035
17	F32420090036	1.30	32420090036
18	F32600020020	5.10	32600020020
19	F36560060055	2.20	36560060055
20	F36560060104	11.60	36560060104
21	F40440080130	10.47	40440080130
22	F40920070168	7.40	40920070168
23	F40940060025	6.80	40940060025
24	F44500060022	14.50	44500060022
25	F44500060057	4.96	44500060057
26	F44500060198	2.29	44500060198
27	F44720070076	1.63	44720070076
28	F44940050325	2.96	44940050325
29	F50600010034	14.00	50600010034
30	F50600010035	17.30	50600010035
31	F50600030153	47.20	50600030153
32	F50600050089	13.40	50600050089
33	F60620050056	3.50	60620050056
34	F60620050199	2.80	60620050199
35	F70860110069	11.90	70860110069
36	F74920080027	17.30	74920080027
37	F76860050185	5.90	76860050185
38	F78460070360	4.25	78460070360
39	F78780020272	51.20	78780020272
40	F80800190010	25.50	80800190010
41	F80800190011	5.80	80800190011

Query executed successfully.

6. Updating values

```
update Land_unit
set unit_cad_value_EUR = 350
where unit_id='44500060198'
```

	Unit_ID	Unit_cad_value_EUR	Prop_ID
14	44500060198	350.00	44500060197

7. Implementing queries

a) Get a list off all properties?

```
select p.prop_id as 'Property ID', p.prop_name as 'Property name' , p.prop_county
as'County', p.prop_parish as 'Parish', sum(t.Type_area_ha) as 'Total area ha'
from property p
inner join Land_unit u
on p.prop_id=u.prop_id
inner join Land_type t
on u.unit_id=t.unit_id
group by p.prop_id, p.prop_name, p.prop_county, p.prop_parish
order by p.prop_name
```

	Property ID	Property name	County	Parish	Total area ha
1	32420090034	Austrini	Aizkraukles	Aiviekstes	11.80
2	60580020248	Bebri	Madonas	Praulienas	6.68
3	32600020020	Bebrupite	Aizkraukles	Kokneses	5.10
4	44500060130	Bez nosaukuma	Augšdaugavas	Demenes	14.50
5	88420080082	Dagas	Siguldas	Allažu	9.30
6	78780020272	Ežmalas	Gulbenes	Jaungulbenes	68.90
7	44940050403	Ganinu mežs	Siguldas	Allažu	2.96
8	88420080043	Gundari	Siguldas	Allažu	7.20
9	44720070076	Janiši-3	Augšdaugavas	Medumu	1.63
10	74920080027	Jauncelmini	Bauskas	Iecavas	17.30
11	40440080131	Jaungrantini	Gulbenes	Jaungulbenes	10.47
12	36560060139	Jaunlici	Siguldas	Mores	2.20
13	96700030090	Jaunmežini	Rezeknes	Comajas	23.30
14	50600020182	Jaunobravieši	Gulbenes	Jaungulbenes	83.00
15	50600010042	Jaunzemi	Gulbenes	Jaungulbenes	17.30
16	40640140210	Klivi	Siguldas	Mores	6.40
17	44500060197	Lakstigalas 2	Siguldas	Allažu	7.26
18	60960020123	Lasenbergi	Kraslavas	Udrišu	11.60
19	78460040396	Lazdini	Augšdaugavas	Demenes	4.25
20	70820070016	Lejas Boksti	Rezeknes	Comajas	20.20
21	36560060140	Lejas Lici	Siguldas	Mores	11.60
22	88820030147	Maravas	Talsu	Rojas	1.70
23	54310030046	Mazperlites	Jelgavas	Kalnciema	7.22
24	56350150027	Mežvidini	Kraslavas	Indras	6.49
25	76860050185	Purvsala	Livanu	Turku	5.90
26	40940060025	Rusa	Bauskas	Vecumnieku	13.86
27	40920070168	Rusas	Bauskas	Iecavas	7.40
28	70860110069	Tursi	Madonas	Praulienas	11.90
29	80800190010	Vecrijnieki	Olaines	Olaines	31.30
30	60620050056	Zalmeži	Kraslavas	Indras	6.30
31	44500060057	Zemgales	Siguldas	Allažu	4.96

b) How many hectares of land does the group own, divided by each land type?

```
select
case
  when left(Type_ID, 1) = 'F' then 'Forest'
  when left(Type_ID, 1) = 'A' then 'Agriculture'
  when left(Type_ID, 1) = 'O' then 'Other'
end as Land_Type_Group,
sum(Type_area_ha) AS Total_Area
from Land_type
group by
case
  when left(Type_ID, 1) = 'F' then 'Forest'
  when left(Type_ID, 1) = 'A' then 'Agriculture'
  when left(Type_ID, 1) = 'O' then 'Other'
end;
```

	Land_Type_Group	Total_Area
1	Agriculture	127.15
2	Forest	301.76
3	Other	11.07

c) How many real estate properties and hectares manages each manager?

```
select m.Mgr_name as 'Name', m.Mgr_surname as 'Surname' ,count(distinct p.prop_id)as
'Number of properties', sum(t.Type_area_ha) as 'Total area ha'
from manager m
inner join company c
on m.mgr_pers_id=c.mgr_pers_id
inner join property p
on c.comp_reg_no=p.comp_reg_no
inner join Land_unit u
on p.prop_id=u.prop_id
inner join Land_type t
on u.unit_id=t.unit_id
group by m.Mgr_name, m.Mgr_surname
order by  sum(t.Type_area_ha)
```

	Name	Surname	Number of properties	Total area ha
1	Juris	Ezers	10	93.78
2	Anna	Berga	7	131.61
3	Andris	Ozols	14	214.59

d) Get a list of land units, where agricultural land is dominating?

```
select p.prop_id as 'Property ID',p.prop_name as 'Property Name',u.unit_id as 'Land unit
ID',
case when left(t.type_id, 1) = 'F' then t.type_area_ha else 0 end as 'Forestland land
ha',
case when left(t.type_id, 1) = 'A' then t.type_area_ha else 0 end as 'Agriculture land
ha',
case when left(t.type_id, 1) = 'O' then t.type_area_ha else 0 end as 'Other land ha',
case
  when max(case when left(t.type_id, 1) = 'A' then t.type_area_ha else 0 end) >
       max(case when left(t.type_id, 1) = 'F' then t.type_area_ha else 0 end) and
       max(case when left(t.type_id, 1) = 'A' then t.type_area_ha else 0 end) >
       max(case when left(t.type_id, 1) = 'O' then t.type_area_ha else 0 end)
  then 'Agriland dominates'
  else 'Agriland does not dominate'
end as 'Agriland dominance'
from property p
inner join land_unit u on p.prop_id = u.prop_id
```

```
inner join land_type t on u.unit_id = t.unit_id
group by p.prop_name, p.prop_id, u.unit_id, t.type_area_ha,t.type_id
having max(case when left(t.type_id, 1) = 'A' then t.type_area_ha else 0 end) >
       max(case when left(t.type_id, 1) = 'F' then t.type_area_ha else 0 end) and
       max(case when left(t.type_id, 1) = 'A' then t.type_area_ha else 0 end) >
       max(case when left(t.type_id, 1) = 'O' then t.type_area_ha else 0 end);
```

	Property ID	Property Name	Land unit ID	Forestland land ha	Agriculture land ha	Other land ha	Agriland dominance
1	40640140210	Klivi	40640140211	0.00	6.40	0.00	Agriland dominates
2	40940060025	Rusa	40940070046	0.00	7.06	0.00	Agriland dominates
3	50600020182	Jaunobravieši	50600010034	0.00	3.00	0.00	Agriland dominates
4	50600020182	Jaunobravieši	50600030153	0.00	2.20	0.00	Agriland dominates
5	54310030046	Mazperlites	54310030114	0.00	7.22	0.00	Agriland dominates
6	56350150027	Mežvidini	56350150027	0.00	6.49	0.00	Agriland dominates
7	60580020248	Bebri	60580020284	0.00	6.68	0.00	Agriland dominates
8	60960020123	Lasenbergi	60960030097	0.00	11.60	0.00	Agriland dominates
9	70820070016	Lejas Boksti	70820070016	0.00	12.50	0.00	Agriland dominates
10	70820070016	Lejas Boksti	70820080009	0.00	2.10	0.00	Agriland dominates
11	70820070016	Lejas Boksti	70820080011	0.00	5.60	0.00	Agriland dominates
12	78780020272	Ežmalas	78780020272	0.00	16.50	0.00	Agriland dominates
13	88420080043	Gundari	88420080043	0.00	7.20	0.00	Agriland dominates
14	88420080082	Dagas	88420080082	0.00	9.30	0.00	Agriland dominates
15	96700030090	Jaunmežini	96700030090	0.00	23.30	0.00	Agriland dominates

e) Get a list of properties, where forestland is dominating?

```
select p.prop_id as 'Property ID', p.prop_name as 'Property Name',
sum(case when left(t.type_id, 1) = 'F' then t.type_area_ha else 0 end) as 'Forestland
land ha',
sum(case when left(t.type_id, 1) = 'A' then t.type_area_ha else 0 end) as 'Agriculture
land ha',
sum(case when left(t.type_id, 1) = 'O' then t.type_area_ha else 0 end) as 'Other land
ha',
case
  when sum(case when left(t.type_id, 1) = 'F' then t.type_area_ha else 0 end) >
    case
      when sum(case when left(t.type_id, 1) = 'A' then t.type_area_ha else 0 end) >
           sum(case when left(t.type_id, 1) = '0' then t.type_area_ha else 0 end)
      then sum(case when left(t.type_id, 1) = 'A' then t.type_area_ha else 0 end)
      else sum(case when left(t.type_id, 1) = 'O' then t.type_area_ha else 0 end)
    end
  then 'Forestland dominates'
  else 'Forestland does not dominate'
end as 'Forestland dominance'
from property p
inner join land_unit u on p.prop_id = u.prop_id
inner join land_type t on u.unit_id = t.unit_id
group by p.prop_name, p.prop_id
having sum(case when left(t.type_id, 1) = 'f' then t.type_area_ha else 0 end) >
case
  when sum(case when left(t.type_id, 1) = 'A' then t.type_area_ha else 0 end) >
       sum(case when left(t.type_id, 1) = 'O' then t.type_area_ha else 0 end)
  then sum(case when left(t.type_id, 1) = 'A' then t.type_area_ha else 0 end)
  else sum(case when left(t.type_id, 1) = 'O' then t.type_area_ha else 0 end)
 end;
```

	Property ID	Property Name	Forestland land ha	Agriculture land ha	Other land ha	Forestland dominance
1	32420090034	Austrini	11.80	0.00	0.00	Forestland dominates
2	32600020020	Bebrupite	5.10	0.00	0.00	Forestland dominates
3	36560060139	Jaunlici	2.20	0.00	0.00	Forestland dominates
4	36560060140	Lejas Lici	11.60	0.00	0.00	Forestland dominates
5	40440080131	Jaungrantini	10.47	0.00	0.00	Forestland dominates
6	40920070168	Rusas	7.40	0.00	0.00	Forestland dominates
7	44500060057	Zemgales	4.96	0.00	0.00	Forestland dominates
8	44500060130	Bez nosaukuma	14.50	0.00	0.00	Forestland dominates
9	44720070076	Janiši-3	1.63	0.00	0.00	Forestland dominates
10	44940050403	Ganinu mežs	2.96	0.00	0.00	Forestland dominates
11	50600010042	Jaunzemi	17.30	0.00	0.00	Forestland dominates
12	50600020182	Jaunobravieši	74.60	5.20	3.20	Forestland dominates
13	60620050056	Zalmeži	6.30	0.00	0.00	Forestland dominates
14	70860110069	Tursi	11.90	0.00	0.00	Forestland dominates
15	74920080027	Jauncelmini	17.30	0.00	0.00	Forestland dominates
16	76860050185	Purvsala	5.90	0.00	0.00	Forestland dominates
17	78460040396	Lazdini	4.25	0.00	0.00	Forestland dominates
18	78780020272	Ežmalas	51.20	16.50	1.20	Forestland dominates
19	80800190010	Vecrijnieki	31.30	0.00	0.00	Forestland dominates

f) How many EUR did each company spend on purchase of real estate in the current and previous year?

```
select c.comp_name as 'Company', year(p.prop_purch_date)as 'Year',
sum(p.prop_purch_value_EUR) as 'EUR'
from company c
inner join property p
on c.comp_reg_no=p.comp_reg_no
where year(p.prop_purch_date)='2024' or year(p.prop_purch_date)='2023'
group by c.comp_name , year(p.prop_purch_date)
order by year(p.prop_purch_date)
```

	Company	Year	EUR
1	Forest, I.K.	2023	252170.00
2	SIA Agro Meži	2023	378700.00
3	SIA ECO FOREST	2023	422635.00
4	SIA FORSH LAND	2023	390000.00
5	Forest, I.K.	2024	341950.00
6	SIA Agro Meži	2024	36125.00
7	SIA ECO FOREST	2024	611990.00
8	SIA FORSH LAND	2024	68400.00

g) Top 10 most expensive purchased properties EUR/ha in current year?

```
select top 10 p.Prop_id as 'Property ID', p.prop_name AS 'Property name',
year(p.prop_purch_date) as 'Purchase year', p.prop_purch_value_EUR as 'Purchase price',
SUM(t.Type_area_ha) as 'Total area ha',
format(cast(p.prop_purch_value_EUR as decimal(10,2)) / nullif(sum(t.Type_area_ha), 0),
'N2') as 'Value EUR/ha'
from property p
inner join Land_unit u on p.prop_id = u.prop_id
inner join Land_type t on u.unit_id = t.unit_id
where year(p.prop_purch_date)='2024'
group by p.prop_id, p.prop_name,p.prop_purch_date, p.prop_purch_value_EUR
order by cast(p.prop_purch_value_EUR as decimal(10,2)) / nullif(sum(t.Type_area_ha),
0)desc;
```

	Property ID	Property name	Purchase year	Purchase price	Total area ha	Value EUR/ha
1	70860110069	Tursi	2024	101150.00	11.90	8,500.00
2	76860050185	Purvsala	2024	50150.00	5.90	8,500.00
3	78460040396	Lazdini	2024	36125.00	4.25	8,500.00
4	78780020272	Ežmalas	2024	585650.00	68.90	8,500.00
5	32600020020	Bebrupite	2024	38250.00	5.10	7,500.00
6	32420090034	Austrini	2024	88500.00	11.80	7,500.00
7	40920070168	Rusas	2024	40700.00	7.40	5,500.00
8	44500060057	Zemgales	2024	21000.00	4.96	4,233.87
9	44940050403	Ganinu mežs	2024	11840.00	2.96	4,000.00
10	44720070076	Janiši-3	2024	6520.00	1.63	4,000.00

Conclusion

The database project successfully achieved its objectives by designing, implementing, and managing a robust and efficient database system. Through a thorough requirements analysis, I developed a comprehensive data model that accurately represented essential entities and their relationships. By applying advanced normalization techniques, I optimized the schema to minimize redundancy and ensure data integrity. The database was implemented using Microsoft SQL Server, enabling seamless storage, retrieval, and management of data.

This project was a transformative learning experience for me. With no prior experience in databases, every aspect was entirely new—from defining business objects to constructing SQL queries. Among the challenges I faced, creating complex queries to identify areas dominated by agriland or forestland proved to be particularly demanding. These queries required significant time and effort, but the process taught me invaluable lessons.

Despite these challenges, the knowledge I gained was remarkable. The hands-on experience of designing the database and implementing queries provided me with a solid foundation in database management. I believe the skills I acquired during this project will continue to benefit me in the future.

References

- Pizika Nadežda – 2023 - Study materials TSI - Modern Database Technologies
- https://www.w3schools.com/
- https://learnsql.com/cookbook/how-to-divide-one-column-by-another-in-sql-server-postgresql-or-sqlite/